Understanding the Macroeconomic Signals Behind the US Dollar Index (DXY)

A Machine Learning Approach to Analyze Direction, Strength, and Lagged Effects of Macroeconomic Indicators

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Research Problem

The DXY Context

US Dollar Index reflects dollar strength against major currencies

The Lag Question

Macroeconomic conditions may influence DXY immediately or with monthly.

Central Question

Which indicators affect DXY, and after how many months does effect appear?

Project Objectives

1

Identify Key Indicators

Determine macroeconomic variables most correlated with DXY movements

2

Analyze Direction

Use linear models to establish positive or negative influence relationships

3

Measure Strength

Apply tree-based models to quantify relative importance of each indicator

4

Examine Timing

Investigate immediate versus lagged effects through temporal analysis

Dataset Overview

Data Structure

- Macroeconomic indicators combined with exchange-data
- Daily frequency aggregated to monthly for lag analysis
- Target variable: DXY_Close
- Time series format supporting temporal modeling

Daily

Original Frequency

Daily observations

Monthly

Aggregated To

Monthly intervals

1

Target Variable

DXY_Close

Data Preparation

01

Raw Data Collection

Combine macroeconomic indicators with exchange-rate data

02

Remove Redundant Prices

Drop Open, High, Low, Adjusted Close retain only DXY_Close

03

Eliminate Multicollinearity

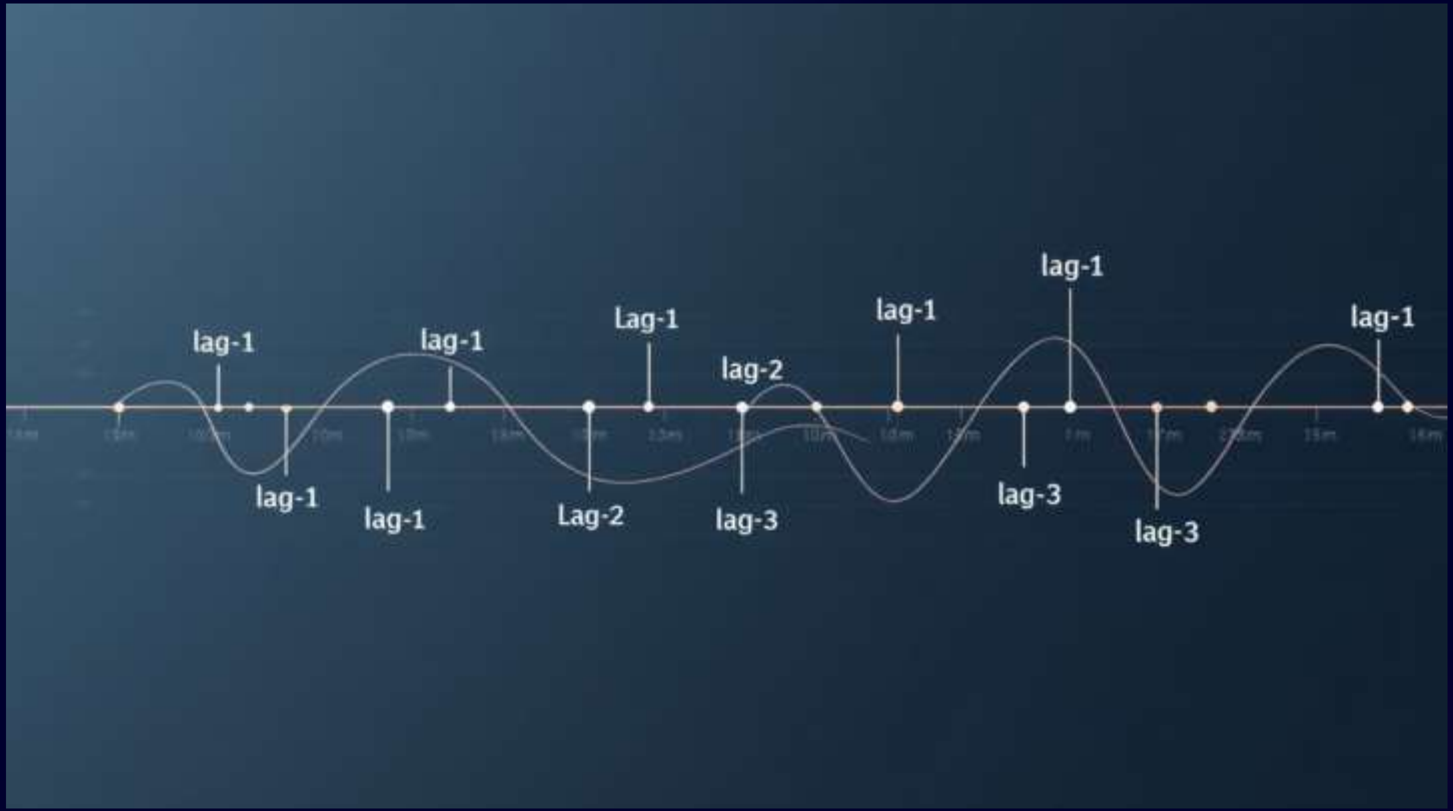
Remove highly correlated variables to reduce redundancy

04

Monthly Aggregation

Transform daily data to monthly frequency for analysis

Feature Engineering



Lag Construction

Create lagged features of 1, 2, and 3 months for macroeconomic indicators

Purpose

Capture delayed economic effects on currency strength

Target Handling

Exclude lagged DXY from analysis to focus on external drivers

1. **Current Month**
   T = 0
2. **Lag 1**
   T - 1 month
3. **Lag 2**
   T - 2 months
4. **Lag 3**
   T - 3 months

Feature Grouping Strategy

Indicators organized into thematic economic blocks for interpretability

Exchange Rate

Currency pairs and cross-rates

Interest Rates

Federal funds, Treasury yields

Inflation

CPI, PPI, inflation expectations

Labor Market

Unemployment, wages, job growth

Credit Conditions

Consumer credit, spreads

Money & Output

M2, GDP, industrial production

Housing

Starts, permits, home prices

Modeling Strategy

Linear Models

- Linear Regression
- Ridge Regression
- ElasticNet

Purpose

Analyze direction and lag timing through coefficient interpretation

Tree-Based Models

- Random Forest

Purpose

Measure relative importance and capture nonlinear relationships

Why Linear Models?

Interpretability

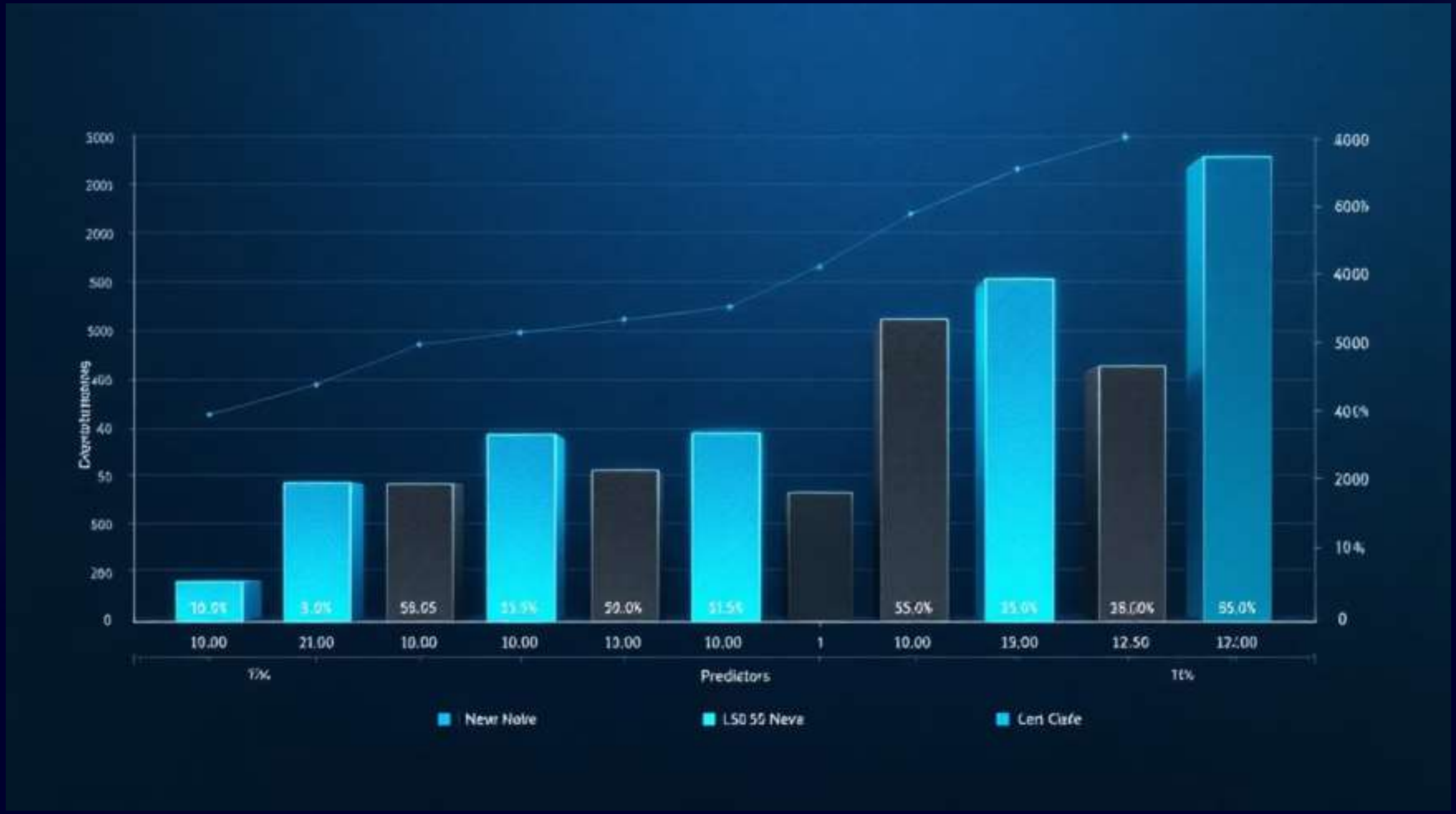
Linear models provide clear, actionable insights

Direction Analysis

Coefficient sign indicates positive or negative relationship with DXY

Timing Identification

Compare current and lagged features to identify strongest effect period



Evaluating Feature Groups Individually

1. **Select Group**

   Choose one economic theme

2. **Include Lags**

   Add lagged variables 1-3 months

3. **Fit Model**

   Apply linear regression

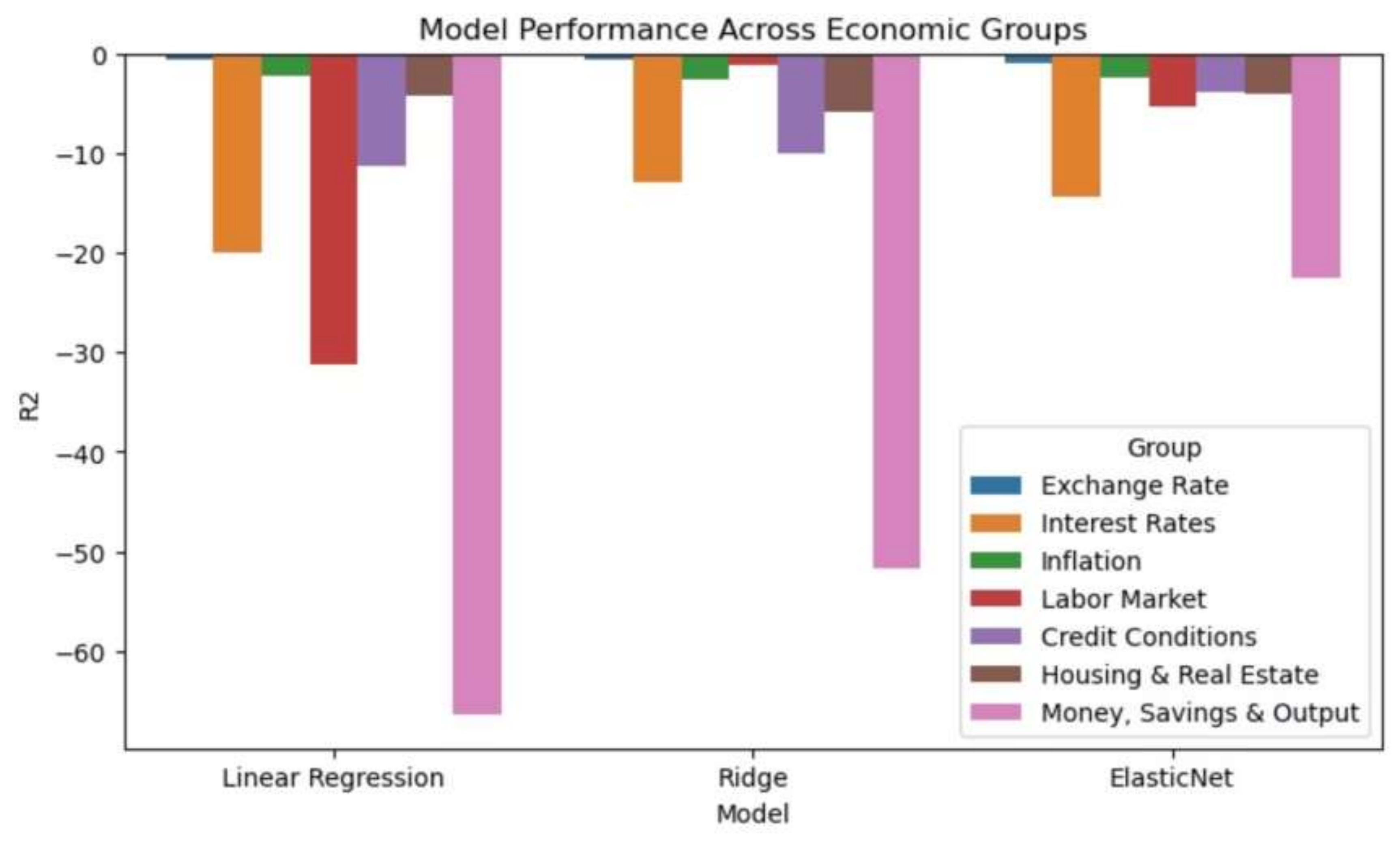
4. **Interpret**

   Analyze directional effects

Linear Model Results by Feature Group

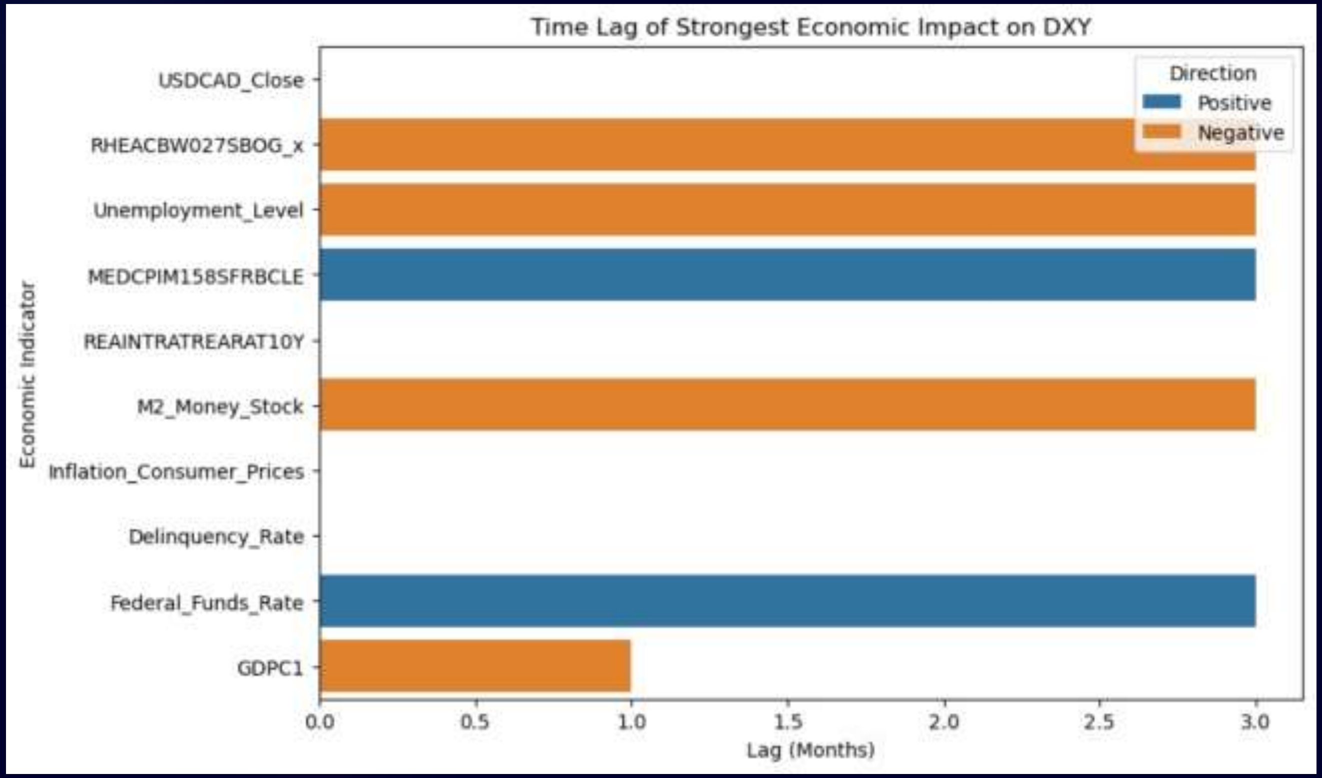
- Most feature groups showed weak explanatory power when tested independently.
- The Exchange Rate group achieved the highest R^2 among all groups.
- ElasticNet gave the best performance across all feature groups.

Conclusion: DXY appears to be driven by multiple interacting factors rather than a single isolated macroeconomic block.



Coefficient Analysis

After how many months does the effect become apparent?



1

Current Month

Some indicators affect DXY immediately in the same period like exchange features and inflation

2

Lag 2+ Months

Longer-term effects materialize after two or more months like Real Gross Domestic Product(GDPC1)

3

Lag 3+ Months

Most of indicators affect after three months like interest features ,labor features and savings

Interpretation of Linear Results

What Linear Models Reveal

- Direction of relationship (positive/negative correlation)
- Timing of delayed effects
- Statistical significance of variables

What Linear Models Do Not Capture

- Nonlinear dependencies
- Complex interactions between variables
- Relative importance ranking

The weak performance of individual feature groups suggests that DXY movements are not driven by one isolated macroeconomic block. Linear models remain valuable for interpretation rather than final feature ranking.

Why Tree-Based Models Were Needed

Linear Models

- Direction and lag analysis
- Timing of effects
- Statistical interpretation

Random Forest

- Relative importance ranking
- Nonlinearity capture
- Interaction effects

Macroeconomic relationships are often nonlinear, and linear models cannot fully capture complex interactions between variables. Tree-based models estimate relative importance more effectively.

Random Forest Model

Model Purpose

Estimate relative importance of macroeconomic features for explanatory analysis

Key Advantage

Captures nonlinear dependencies and complex interactions between variables

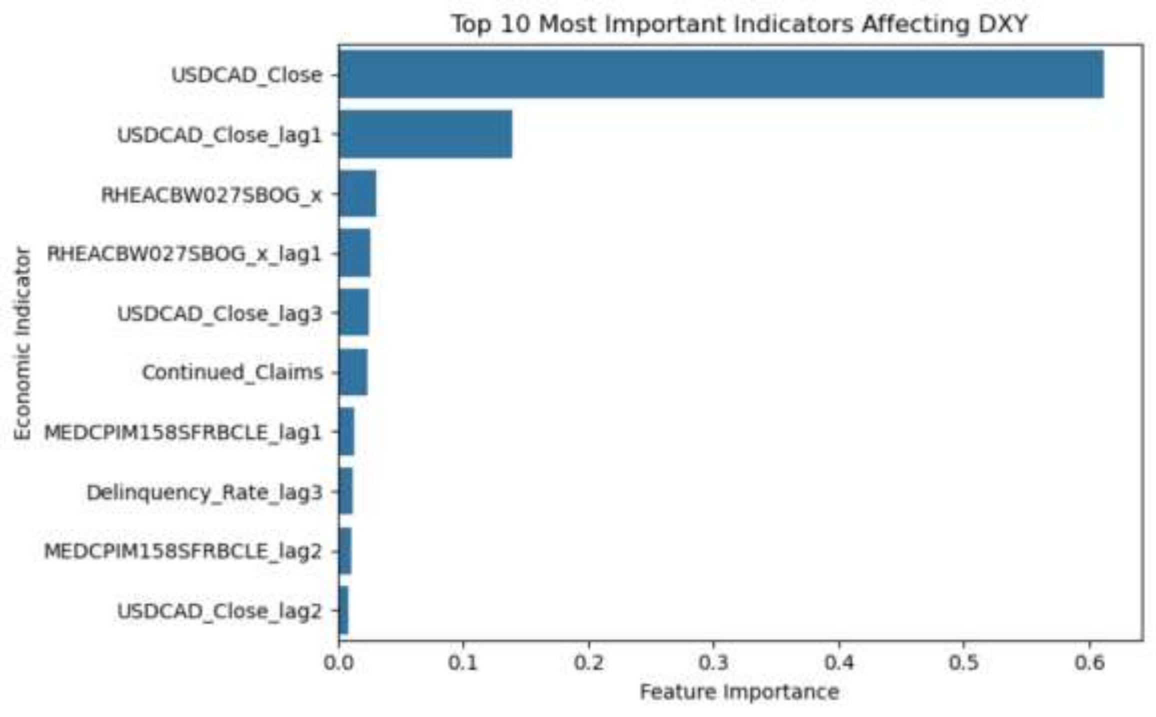
Application

Explanatory analysis rather than prediction accuracy alone

0.0s

	Feature	Importance
0	USDCAD_Close	0.611737
14	USDCAD_Close_lag1	0.138780
12	RHEACBW027SBOG_x	0.030937
50	RHEACBW027SBOG_x_lag1	0.025139
16	USDCAD_Close_lag3	0.024877
3	Continued_Claims	0.022789
44	MEDCPIM158SFRBCLE_lag1	0.012543
28	Delinquency_Rate_lag3	0.012151
45	MEDCPIM158SFRBCLE_lag2	0.011000
15	USDCAD_Close_lag2	0.008440
51	RHEACBW027SBOG_x_lag2	0.007098
46	MEDCPIM158SFRBCLE_lag3	0.006503
49	PSAVERT_lag3	0.006402
39	M2_Money_Stock_lag2	0.005967
38	M2_Money_Stock_lag1	0.005207

Feature Importance



Exchange-rate indicators dominated the importance ranking, with USDCAD emerging as the most influential feature.

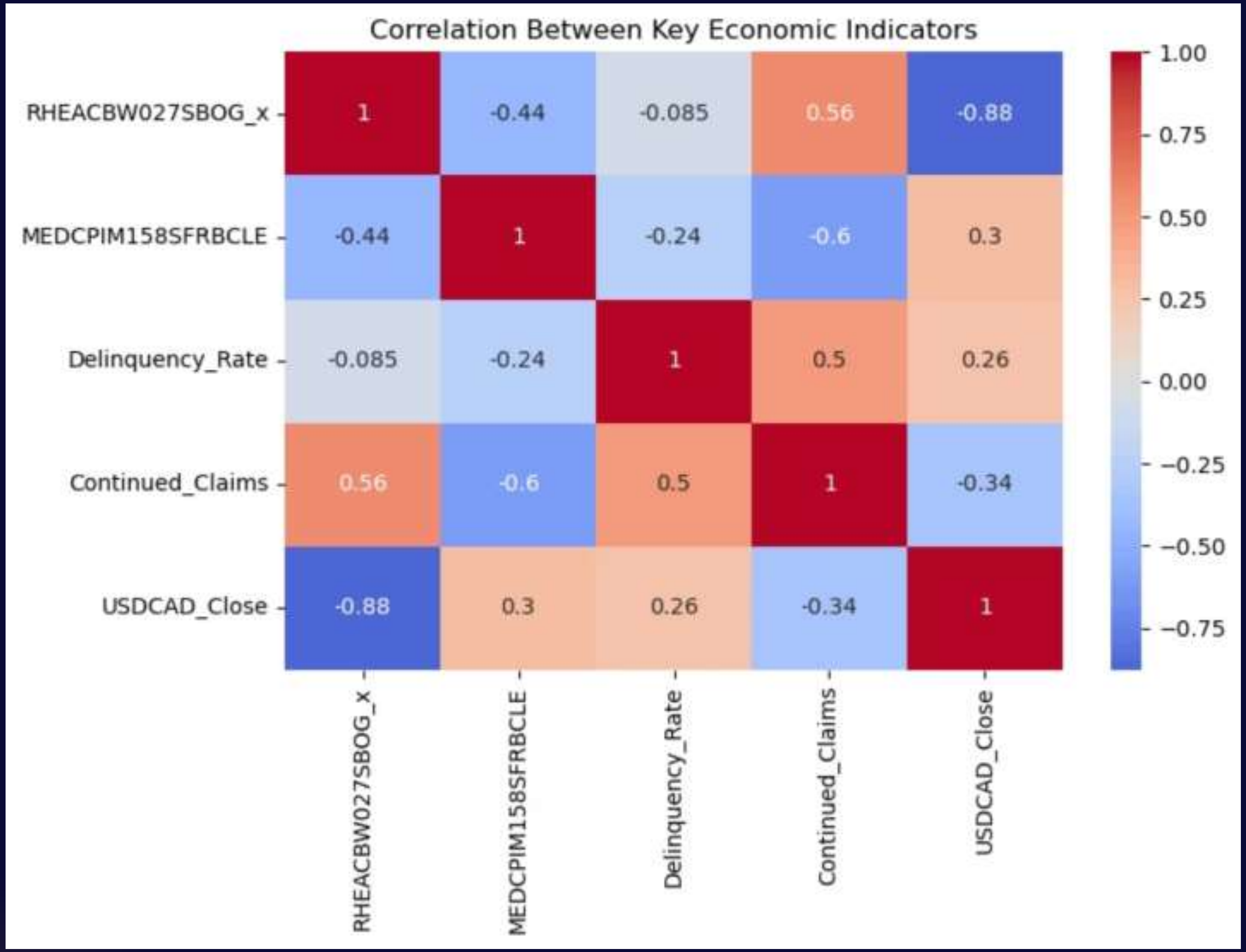
Other macroeconomic indicators contributed with smaller but still meaningful effects, providing additional context for DXY movements.

Strongest Correlation with DXY

Correlation Analysis Insights

Correlation analysis supports the broader interpretation of feature-importance results. Exchange-rate variables show the strongest relationship with DXY, while inflation and interest-rate variables show moderate relationships.

The correlation perspective validates the machine learning findings, providing complementary evidence for the dominant role of currency exchange rates.



Key Findings

Exchange-Rate Dominance

Exchange-rate indicators had the strongest relationship with DXY

USDCAD Lead

USDCAD emerged as the most influential variable in tree-based analysis

Supporting Signals

Inflation and interest-rate indicators provided additional but weaker signals

Timing Variation

Some effects appeared immediately, while others emerged after delay

Multiple Factors

DXY influenced by multiple factors rather than single macroeconomic block

What we learned

- We learned that **model accuracy is not always the most important goal**, especially in explanatory projects like this one. our case, interpretation and economic insight were more valuable than simply maximizing predictive performance.
- We learned how to use **linear models to interpret direction and lag effects**, and how tree-based models can provide a clearer view of **relative feature importance**.
- The project helped us understand that **most macroeconomic indicators cannot explain DXY independently**; the dollar index is shaped by multiple interacting factors rather than a single isolated variable.
- We also gained practical knowledge in **macroeconomics and financial indicators**, which deepened our understanding of of the economic meaning behind the data, not only the machine learning side.
- Finally , We tried several additional experiments, such as predicting **DXY return instead of DXY close**, we learned more about the mathematical behavior of the models and the economic structure of the indicators, even when these experiments were not included in the final scope due to time constraints.

Project Timeline & Milestones

1

Weeks 1-3

Data collection from Federal Reserve, cleaning, and monthly conversion and Lag features engineering and economic block grouping.

2

Weeks 4

Baseline linear models and explanatory performance analysis

3

Weeks 5

Random Forest implementation for nonlinear relationships and feature importance

4

Week 6

Correlation analysis, interpretation, and final report and presentation completion.

Future Work

Three promising directions to extend this DXY forecasting framework

Advanced ML Architectures

LightGBM, MLP, LSTM, GRU networks to capture nonlinear dependencies

Joint Indicator Modeling

Simultaneous analysis of macroeconomic variable interactions shaping DXY movements

Time Series Forecasting

Alternative targets including DXY returns and extended lag structures

Thank You

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